JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/02

Paper 2 Core Paper **2 September 2015**

Additional Materials: Answer Paper **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question.
Circle the question number of the question attempted.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
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Section B	
9 / 10	
Total	

This document consists of **22** printed pages and **4** blank pages.

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Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows replication of a part of the glucagon receptor gene.

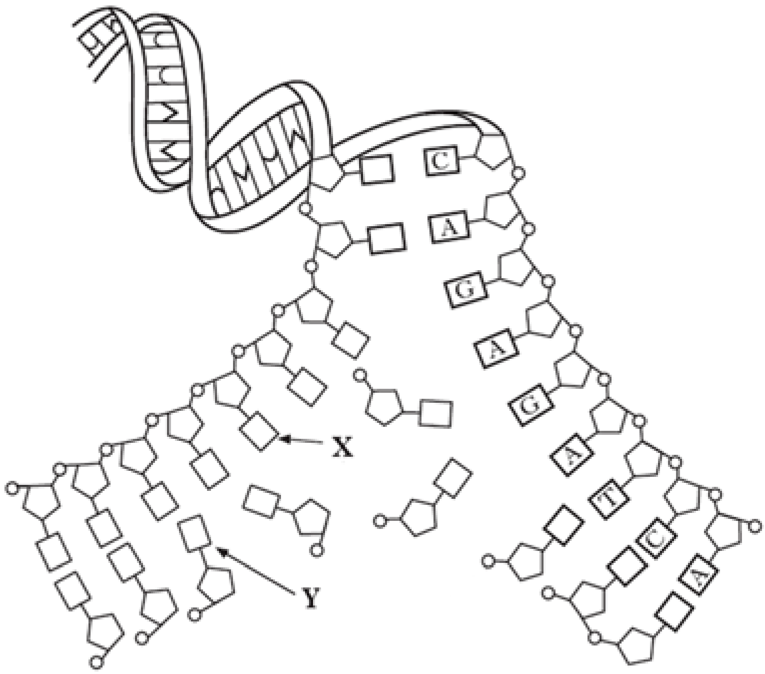

Fig. 1.1

(a) (i) Name the bases labelled X and Y on Fig. 1.1. [1]

X ______________________________

Y ______________________________

(ii) State the phase of the cell cycle at which DNA replication occurs. [1]

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(iii) DNA replication involves a number of enzymes including DNA polymerase. Identify two other enzymes involved in DNA replication. [2]

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(iv) Explain how Fig. 1.1 shows semi-conservative replication. [2]

The structure of a glucagon receptor is shown in Fig. 1.2.

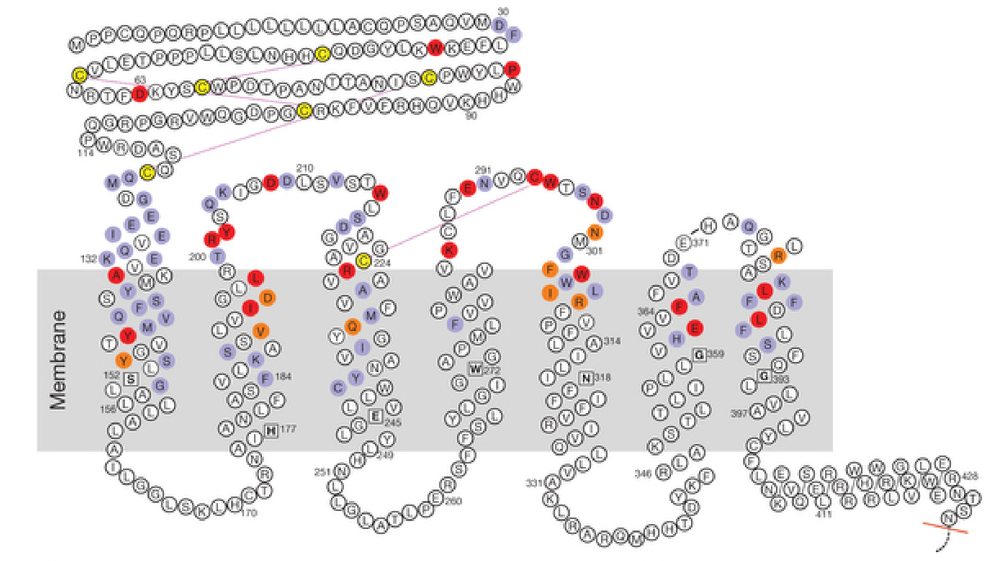

Fig. 1.2

(b) Describe how amino acid residues at different positions in the polypeptide may be brought together in the receptor site when the glucagon receptor is synthesized. [4]

In 2-8% of white populations, a mutation was incorporated in exon 2 of the glucagon receptor gene, which changes a glycine to a serine resulting in a Gly40Ser mutant receptor. The Gly40Ser mutant receptor was found to bind glucagon with an approximately three-fold lower affinity compared with the wild type receptor. This mutation located in the extracellular region of the glucagon receptor decreases the sensitivity of target tissues to glucagon.

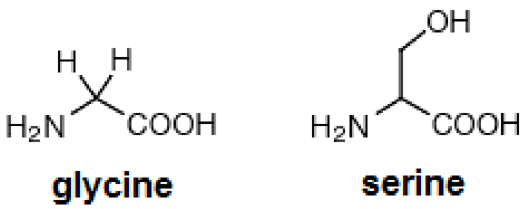

Fig. 1.3

(c) With reference to the information provided and Fig. 1.3, describe the effects of the mutation of the glucagon receptor gene. [4]

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[Total: 14]

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2 Fig. 2.1 shows the main structural features of a mature (infective) human immunodeficiency virus (HIV).

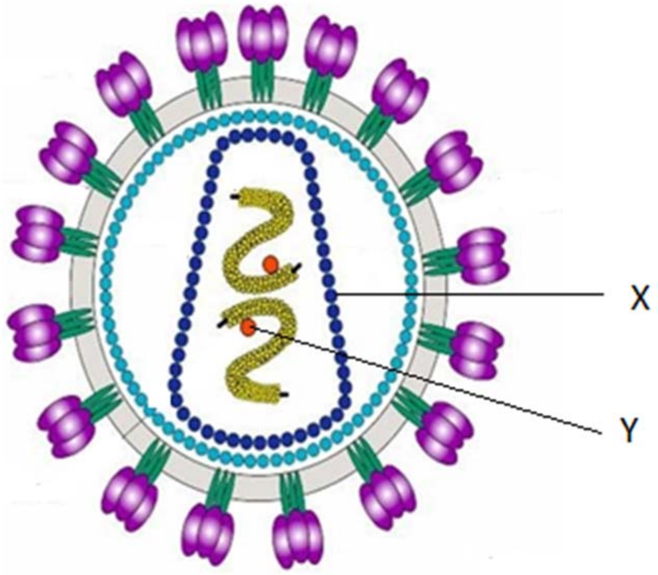

Fig. 2.1

(a) With reference to Fig. 2.1, label molecules X and Y. [2]

X ____

Y ____

(b) Describe the roles of HIV RNA in the infection process of T cells. [2]

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(c) Raltegravir has been the first and is currently the only integrase strand transfer inhibitor (InSTI) approved for the treatment of HIV. Raltegravir inhibits integrase by binding to the active site and sequestering essential active site magnesium ions and the HIV genes are not expressed into HIV enzymes.

Suggest how Raltegravir work in the treatment of HIV infection. [2]

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Fig. 2.2 shows the structure of Middle East respiratory syndrome coronavirus (MERS-CoV). MERS affects the respiratory system (lungs and breathing tubes). Most MERS patients developed severe acute respiratory illness with symptoms of fever, cough and shortness of breath. About 3-4 out of every 10 patients reported with MERS have died.

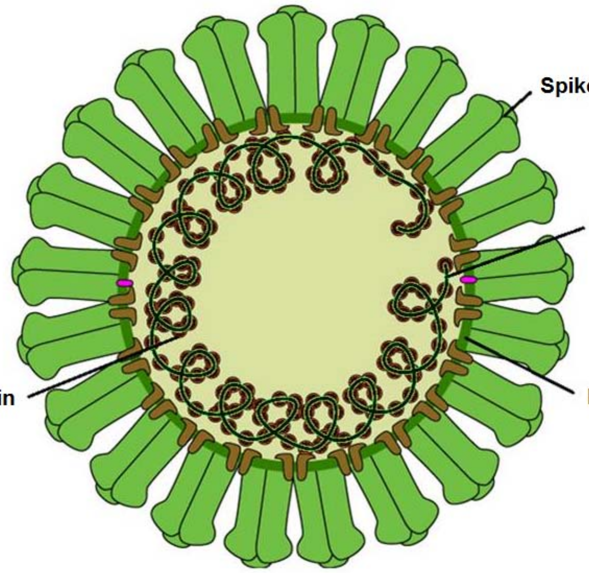

Fig. 2.2

(d) (i) With reference to Fig. 2.1 and Fig. 2.2, compare the structure of HIV and MERS-CoV. [2]

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(ii) The evidence available to date suggests that the coronavirus have been present in bats for some time and had spread to camels by the mid-1990s. The viruses appear to have spread from camels to humans in the early 2010s.

What changes may have occurred to give MERS-CoV the ability to infect humans? [2]

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[Total: 10]

3 A recent study investigated a chromosome mutation discovered in malignant lung tumour cells. The mutation results in the fusion of the two genes shown in Fig. 3.1. The genes are *EML4*, which is involved in microtubule formation, and *ALK* which codes for a kinase enzyme. As a result of this fusion, a new protein (EML4-ALK) is formed that has uncontrolled kinase activity inside the cell. in many cases, the EML4-ALK protein promotes and maintains the malignant behaviour of the cancer cells.

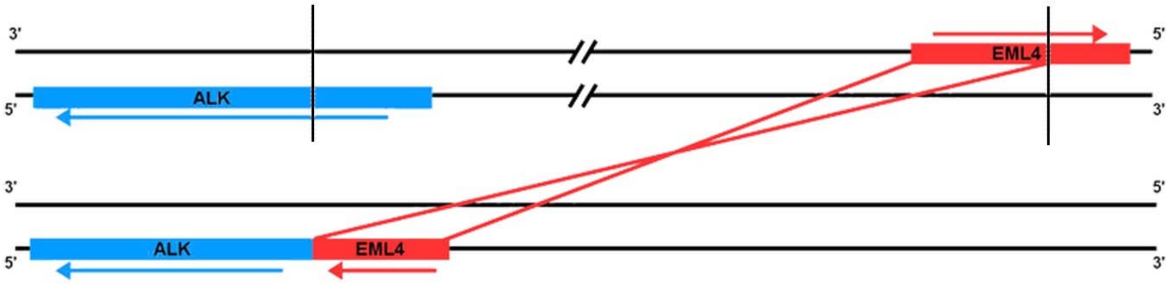

Fig. 3.1

(a) With reference to Fig. 3.1, describe how the proto-oncogene may be converted into an oncogene. [1]

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Part of the study aimed to find out if normal cells are transformed to divide abnormally after treatment with the fusion gene. In the treatments, genes were introduced into normal cells that were then cultured in a flat dish containing a suitable growth medium. Abnormal cell division is indicated by the transformed cells stacking up in multiple layers called *foci*.

The four gene treatments used and the results of the cell transformation study are shown in the Table 3.1. In the fourth treatment, the *EML4–ALK* fusion gene was modified so that the kinase produced was inactive.

Table 3.1

gene treatment	formation of foci
ALK alone	no
EML4 alone	no
EML4-ALK fusion	yes
EML4-ALK fusion modified	no

(b) Explain why normal cells were observed to be in monolayer before *EML4–ALK* fusion gene was introduced. [1]

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(c) With reference to Table 3.1, explain how the results show that the mutation created an oncogene. [1]

(d) In liver cancer, *EML4–ALK* fusion gene is only present in some liver cells.

Explain why not all cells in the patient's liver are transformed into cancer cells. [3]

[Total: 6]

4 In fruit flies, *Drosophila melanogaster*, the genes for body colour and wing length are at separate loci. A scientist investigated the relationship between the two genes. He carried out crosses between fruit flies with grey bodies and long wings and fruit flies with black bodies and short wings.

Fig. 4.1 shows his crosses and the result.

parental phenotypes:	**grey body, long wings**	**X**	**black body, short wings**
offspring phenotype :	**All grey body, long wings**		

These offspring were crossed with flies homozygous for black body and short wings.

Fig. 4.1

The phenotypes of the offspring of the test cross were recorded and shown in Table 4.1.

Table 4.1

phenotype	observed number
grey body, long wings	113
black body, short wings	120
grey body, short wings	15
black body, long wings	16

(a) Draw a genetic diagram to explain the results shown in Table 4.1. [3]

Use the following symbols to represent the different alleles involved:

G - grey body   **g** black body
N - long wings   **n** short wings

(b) Use your knowledge of gene linkage to explain these results. [3]

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(c) If these genes were not linked, what ratio of phenotypes would the scientist have expected to obtain in the offspring? [1]

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The equation to the chi-squared test and part of the critical values of the chi-squared distribution is shown below.

$$\chi^2 = \Sigma \frac{(O - E)^2}{E} \qquad\qquad v = c - 1$$

Key to symbols

Σ = 'sum of' $\qquad$ v = number of degrees of freedom

O = observed 'value' $\qquad$ c = number of classes

E = expected 'value'

	Probability								
df	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750

(d) Using the chi-squared distribution, apply the chi-squared test to determine whether the results obtained conformed to the expected Mendelian ratio for such a cross. [4]

[Total: 11]

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5 The plasma membrane of beetroot cells contains protein and phospholipids and has a porous nature. As a result, the membrane is selectively permeable.

Cyanide is a poison that inhibits cytochrome c oxidase involved in aerobic respiration.

Fig. 5.1 shows how cyanide concentration affects the uptake of chloride ions by beetroot cells. The rates of chloride ion uptake are given as percentages of those obtained in a control experiment with no cyanide.

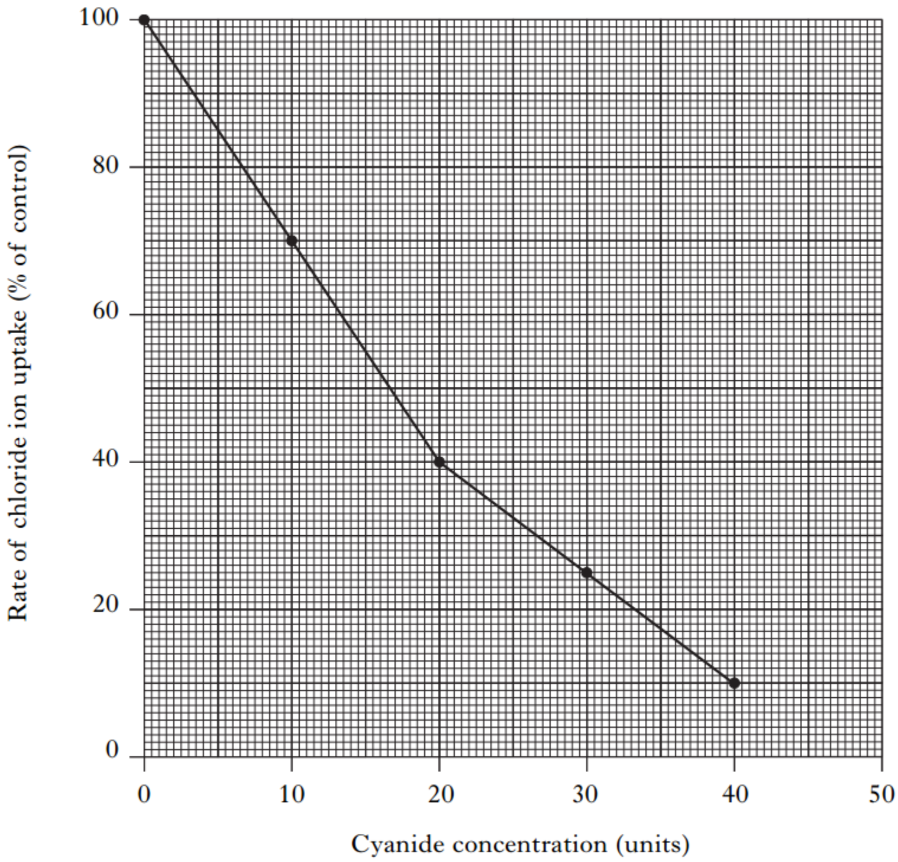

Fig. 5.1

(a) **(i)** It is known that the uptake of chloride ions requires the presence of transmembrane protein.

Explain why chloride ions cannot freely pass through the phospholipid bilayer of membranes. [2]

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(b) With reference to Fig. 5.1,

(i) identify how chloride ions would be moved across the membrane, [1]

(ii) account for your answer in (b)(i). [3]

(c) The activity of cytochrome c oxidase was measured at different values of pH by using nine different buffer solutions. The temperature was kept constant at 30°C.

The results are shown in Fig. 5.2.

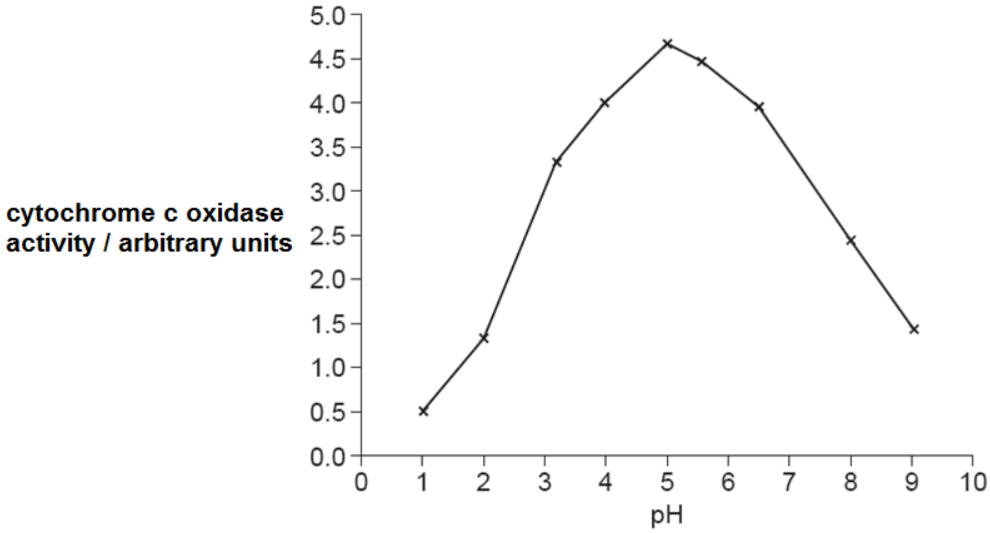

Fig. 5.2

(i) With reference to Fig. 5.2, describe the effect of pH on the activity of cytochrome c oxidase. [2]

(ii) Explain why the activity of cytochrome c oxidase at pH 1 is very low. [3]

[Total: 11]

6 Fig. 6.1 shows a normal mitochondrion and a mitochondrion with a structural defect due to lethal cell injury.

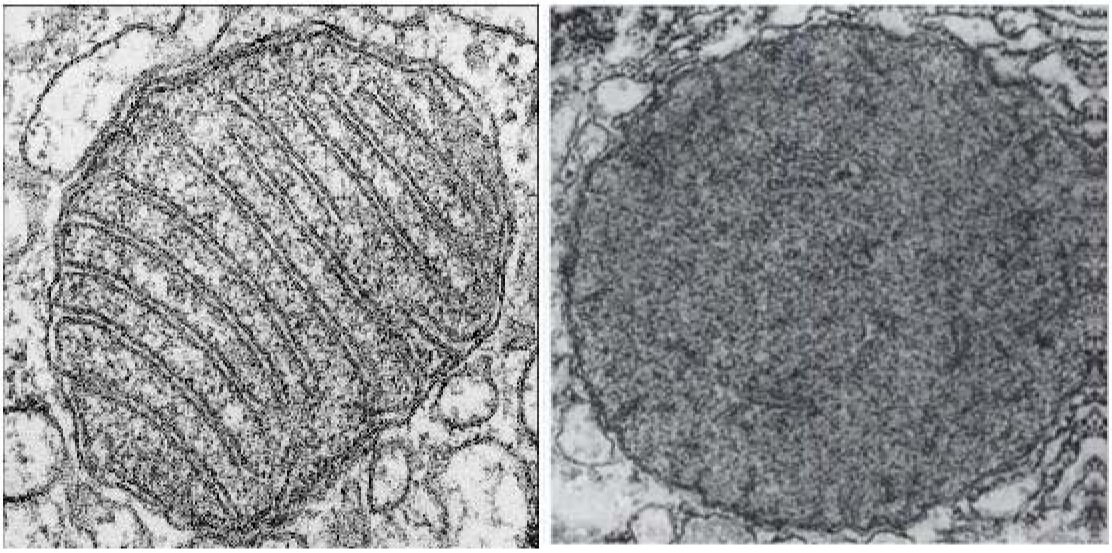

Fig. 6.1

(a) State the mitochondrial structural defect shown in Fig. 6.1. [1]

(b) Explain the implications of this structural defect on

(i) glycolysis, [2]

(ii) oxidative phosphorylation. [2]

(c) The metabolic pathway in which a hexose sugar, such as glucose, is broken down in respiration by cells starts with glycolysis as shown in Fig. 6.2.

The enzymes responsible for catalyzing the first and last key steps, phosphofructokinase (PFK) and pyruvate kinase are the primary steps for allosteric enzyme regulation. PFK is allosterically inhibited by ATP, so glycolysis is slowed when cellular ATP concentrations are high.

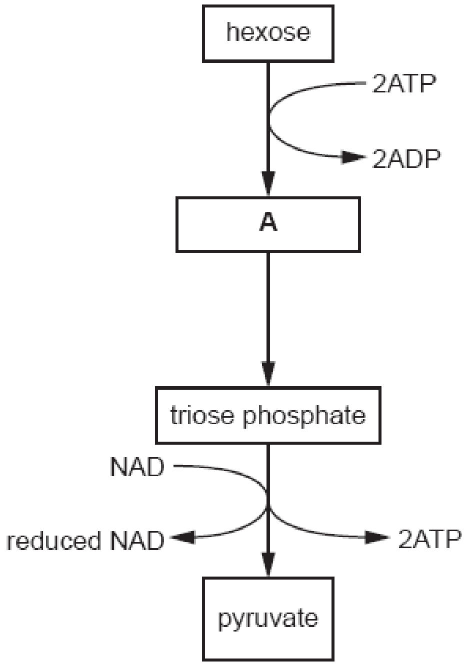

Fig. 6.2

(i) Name substance A and explain why hexose is converted to substance A. [2]

(ii) Explain the role of NAD in aerobic respiration. [2]

[Total: 9]

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7 An investigation was carried out into the effect of eating two different sources of carbohydrate on the concentration of glucose and the concentration of insulin in the blood plasma. The carbohydrate sources used were glucose and rice. The carbohydrate in rice is mainly in the form of starch.

Volunteers were fed either 50 g of glucose or the quantity of rice known to contain 50 g of carbohydrate. The concentration of glucose and insulin in their blood plasma was then measured at intervals for 3 hours. The results are shown in Fig. 7.1 and Fig. 7.2.

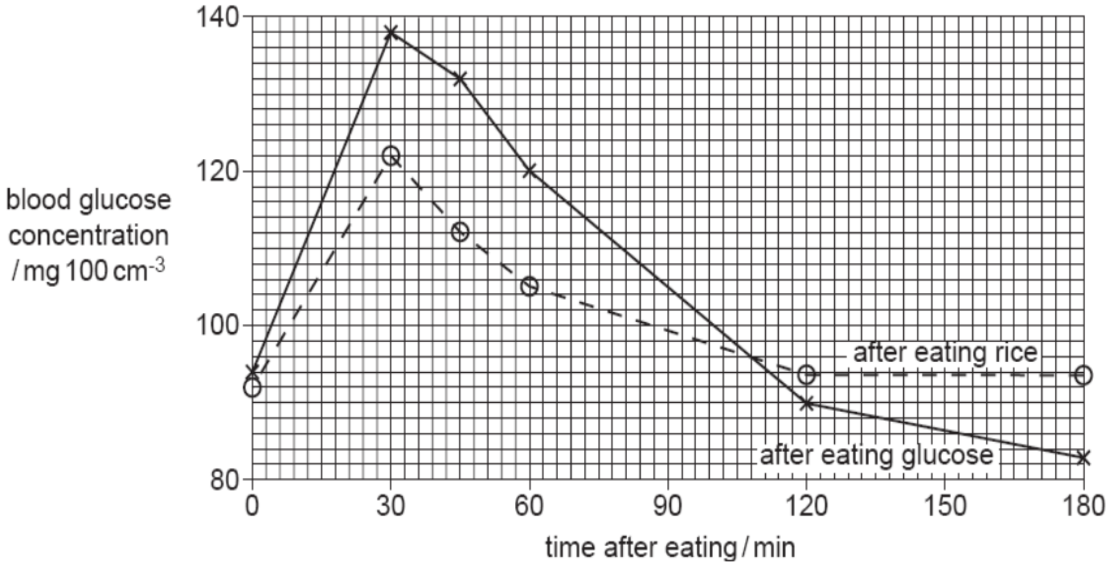

Fig. 7.1

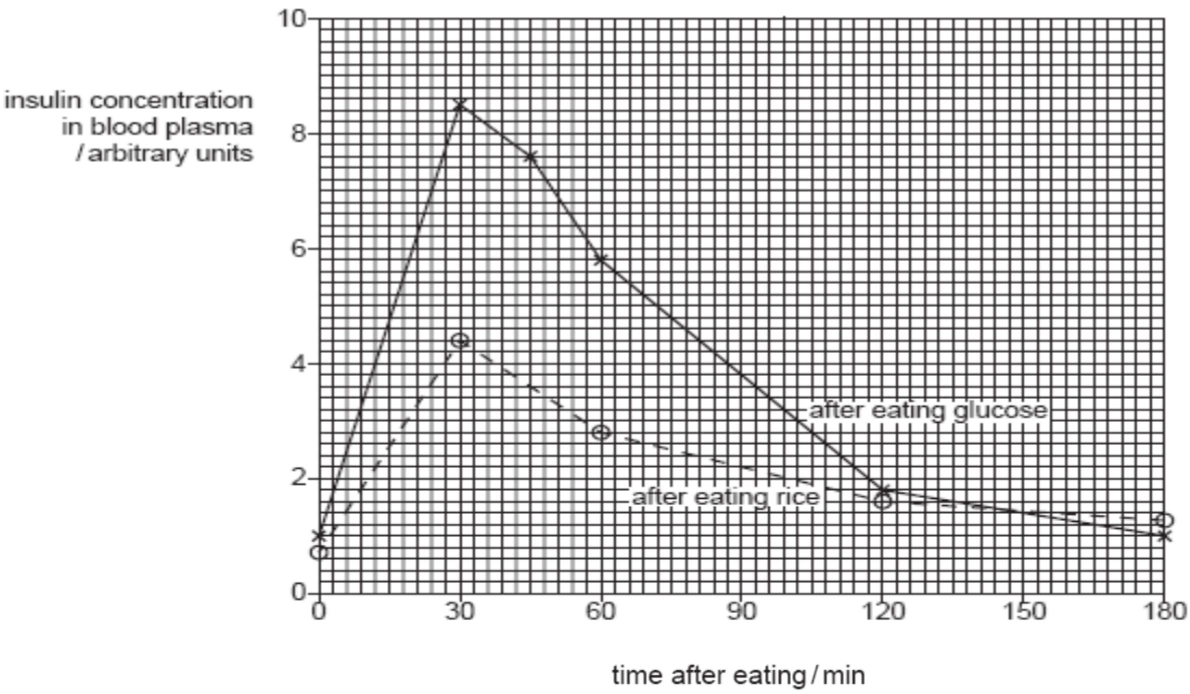

Fig. 7.2

(a) With reference to Fig. 7.1, give an explanation for the difference in concentrations of blood glucose in the first 30 minutes after eating glucose and after eating rice. [1]

(b) With reference to Fig. 7.1 and Fig. 7.2, explain the differences between the concentration of insulin in the blood at 30 minutes after eating glucose and after eating rice. [4]

(c) By 180 minutes after eating the glucose or rice, the concentration of glucose in the blood had fallen considerably.

Describe the role of skeletal muscle in bringing about the reduction in blood glucose concentration. [1]

Fig. 7.3 shows insulin activation of a GLUT4 transporter that aids in removing excess glucose from the blood.

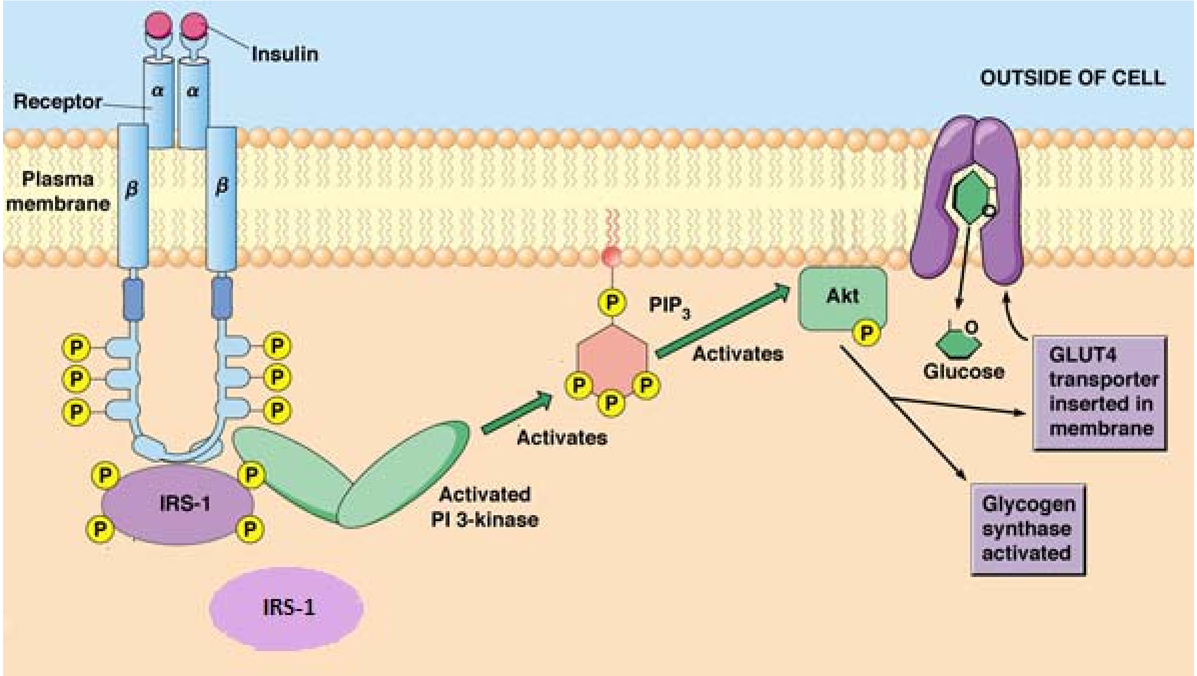

Fig. 7.3

(d) (i) With reference to Fig. 7.3, describe two advantages of this cell signalling pathway. [2]

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(ii) Explain how GLUT4 transporter could remain embedded in the plasma membrane. [1]

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[Total: 9]

8 There are over 40 Galapagos Islands including the small and isolated island named Daphne Major.

Studies were made every year from 1970 to 1989 on the beak size of the island's population of ground finch, *Geospiza fortis*, by measuring the beak length of every bird (Fig. 8.1). Larger finches with larger beaks are better at opening large seeds.

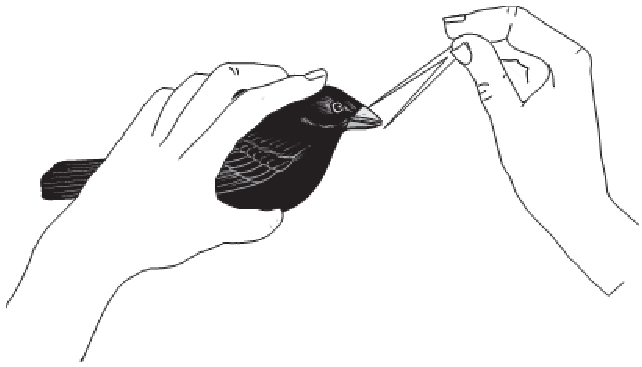

Fig. 8.1

From 1976 to 1978 there was a drought and only 15% of the ground finches survived and these did not breed during drought years. The most conspicuous feature of the survivors of the drought years was their large beak size.

During normal years, many drought intolerant grasses and herbs produce an abundance of small seeds. A few other drought-tolerant plants produce a much smaller number of large seeds which are not normally eaten.

(a) Describe how environmental factors act as forces of natural selection, during drought years, on the beak size of finches. [4]

(b) Suggest the role of the islands in the evolution of thirteen species of Darwin finches now found on the Galapagos Islands. [2]

Molecular analysis was carried out on the mitochondrial DNA (mtDNA) sequences of the Galapagos islands finches and the Cocos finch found on the island of Cocos, 830 km to north-east of the Galapagos islands. Using mtDNA analysis data, a map showing the phylogeny of these finches was constructed as shown below in Fig. 8.2

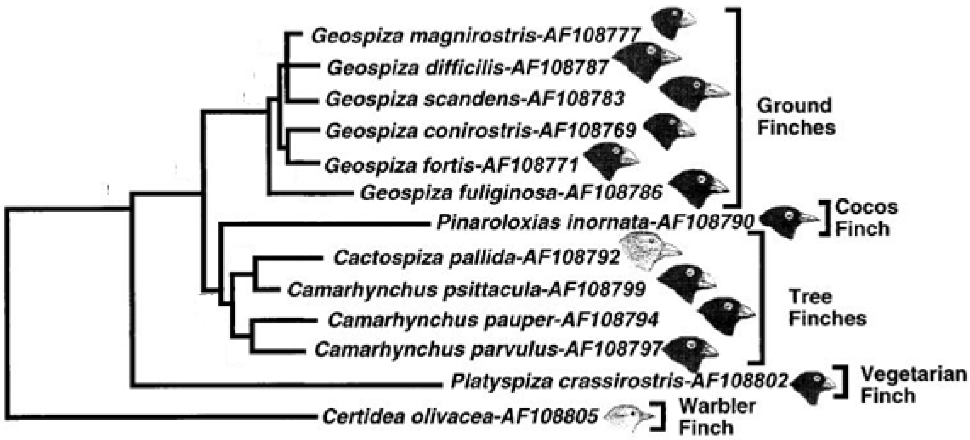

Fig. 8.2

(c) Explain how DNA sequences can be used to determine evolutionary relatedness between species. [2]

(d) Describe the advantages of using nucleotide data such as mtDNA in classifying organisms. [2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Outline how the structure of a named structural carbohydrate is related to its specific functions. [7]

(b) Compare and contrast collagen and haemoglobin in terms of their structure and functions. [6]

(c) Discuss the roles of components of telomerase and telomeres in enabling the function of stem cells. [7]

[20]

10 **(a)** Describe the behaviour of chromosomes during the mitotic cell cycle. [8]

(b) Explain the importance of mitosis in growth, repair and asexual reproduction. [6]

(c) Describe the differences between continuous and discontinuous variation. [6]

[20]